Master List of Geospatial & Climate Data Sources

A comprehensive, verified directory of free, open-access, and authoritative platforms hosting environmental datasets for machine learning pipeline developments.

🌍 1. Digital Elevation & Terrain Data (DEM)

NASA Earthdata (ASTER GDEM / NASADEM)

https://earthdata.nasa.gov/

High-resolution global digital elevation maps at 30-meter precision. Perfect for calculating slope angles and defining watershed boundaries.

USGS EarthExplorer

https://earthexplorer.usgs.gov/

Global SRTM (Shuttle Radar Topography Mission) radar data arrays and historical planetary aerial mapping datasets.

OpenTopography

https://opentopography.org/

A specialized academic repository offering open access to high-resolution lidar topography point-cloud frameworks globally.

🛰️ 2. Satellite Imagery & Cloud Cover Data

Copernicus Browser (Sentinel Hub)

https://browser.dataspace.copernicus.eu/

Direct interface for Sentinel-2 (10m optical) and Sentinel-1 (Radar/SAR). Synthetic Aperture Radar handles cloud penetration to map real-time surface ground moisture variations.

NASA LAADS DAAC (MODIS & VIIRS)

https://ladsweb.modaps.eosdis.nasa.gov/

Daily wide-swathe global satellite passes. Optimized for monitoring daily changes in large-scale snowmelt patterns and convective cloud movements.

USGS Landsat (via EarthExplorer)

https://earthexplorer.usgs.gov/

Decades-long multispectral optical and thermal data streams captured at 30-meter pixel resolution for long-term environmental trends.

🌦️ 3. Historical Weather Reanalysis & Climate Data

ECMWF Copernicus Climate Data Store (ERA5)

https://cds.climate.copernicus.eu/

The absolute gold standard for weather-based AI training datasets. Delivers continuous hourly estimations of global surface temperature, air pressure metrics, humidity indexes, and total precipitation from 1940 down to modern runtime.

NOAA Physical Sciences Laboratory (PSL)

https://psl.noaa.gov/data/

Free access to historical gridded climate logs, specialized gauge-based daily rainfall metrics, and global upper-air atmospheric soundings.

NASA MERRA-2

https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/

Satellite-era global climate reanalysis model highlighting complex atmospheric composition, aerosol behaviors, and hydrological water-cycle states.

📡 4. Real-Time Weather APIs

Open-Meteo API

https://open-meteo.com/

Production-grade weather ingestion engine completely free for developer usage without API keys. Delivers clear, raw JSON responses matching historical ERA5 structures.

OpenWeatherMap (One Call API)

https://openweathermap.org/

Developer tier offering 1,000 requests per day for live situational updates regarding humidity shifts, barometric trend fluctuations, and 1-hour rainfall accumulation layers.

WeatherAPI

https://www.weatherapi.com/

Free operational feed layer granting instant lookups for active ambient indicators, reverse geocoding data, and trailing 7-day historical slices.

💻 5. Multi-Dataset Integrators (Cloud Catalogs)

Google Earth Engine (GEE)

https://earthengine.google.com/

Massive multi-petabyte public environmental data warehouse containing native, pre-processed collections of ERA5, Sentinel, and NASADEM assets manageable directly via Python.

Microsoft Planetary Computer

https://planetarycomputer.microsoft.com/

Cloud-optimized spatial platform utilizing standardized STAC querying configurations to deliver highly efficient cloud-native machine learning datasets.

Amazon Web Services (AWS) Open Data Registry

https://registry.opendata.aws/

Hosts zero-friction, public cloud object-storage buckets featuring critical heavy datasets such as live NOAA NEXRAD weather radar telemetry streams.

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